GREEN ALIBI: Does Fluorescence Catch Drought Stress Before NDVI?

Marathwada, Maharashtra, India | 8 districts | 9 Kharif seasons (2015–2023) | TROPOMI cross-check

STUDY
Agricultural drought stress
across 8 districts

→

SATELLITE + RAINFALL
GOSIF SIF | MODIS NDVI
TROPOMI SIF | CHIRPS

→

SEASONAL LEAD
Two normalisations
Cross-correlation
40-variant method check

→

DROUGHT SIGNAL
Anomaly vs normal years
Region + 72 district-years

→

POLICY CHECK
Rainfall Trigger-1
2015 & 2018 declarations

KEY RESULTS

NOT AN EARLIER ALARM
Drought anomaly ($z < -3$): 2015 same day,
2018 NDVI 8 days before SIF
Districts: same 6 | NDVI first 5 | SIF first 1

SCALING-DEPENDENT LEAD
Season-scaled: SIF first 9/9 years
Decline-scaled: SIF first 6/9 years
Cross-corr.: 6 SIF | 1 tie | 2 NDVI

NO DISTRICT EFFECT
District differences vanish when
curves are scaled over their own decline
TROPOMI: SIF first 3/3 years (2018–2020)

2018 DROUGHT TIMELINE
alarms 35.5 (NDVI) and 27.5 (SIF) days before declaration

CONCLUSION • SIF is not an earlier drought alarm than NDVI; its seasonal lead is largely a scaling effect.